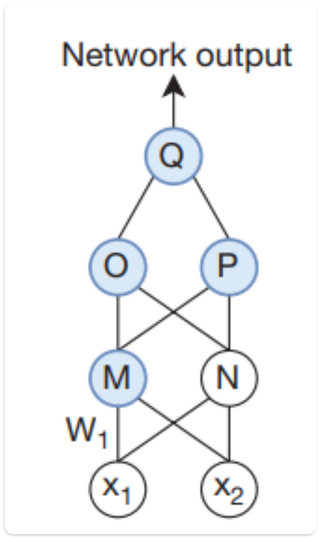
1. Backpropagation for More Complex Neuron Network—Example 1:

A.



B. Error M:

$$\Rightarrow Error\,Q = \frac{\partial e}{\partial z_Q} = \frac{\partial e}{\partial f_Q}\cdot\frac{\partial f_Q}{\partial z_Q}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = \frac{\partial e}{\partial f_Q}\cdot\frac{\partial f_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = Error\,Q\cdot W_{OQ}\cdot y_O'(z_O)$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = \frac{\partial e}{\partial f_Q}\cdot\frac{\partial f_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = Error\,Q\cdot W_{PQ}\cdot y_P'(z_P)$$

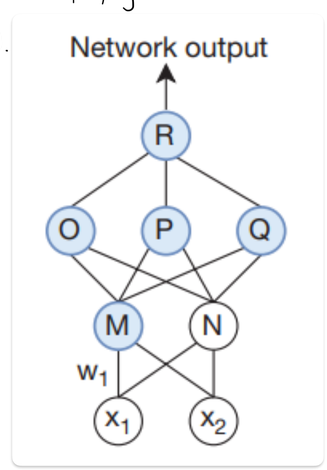
$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = \frac{\partial e}{\partial f_Q}\cdot\frac{\partial f_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}\cdot\frac{\partial z_O}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_Q}\cdot\frac{\partial f_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}\cdot\frac{\partial z_P}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M}$$

$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = Error\,O\cdot W_{MO}\cdot f_M'(z_M) + Error\,P\cdot W_{MP}\cdot f_M'(z_M) = (Error\,O\cdot W_{MO} + Error\,P\cdot W_{MP})\cdot f_M'(z_M)$$

C. $W_1' = W_1 + \eta\cdot(-1\cdot x_1\cdot Error\,M)$

2. Backpropagation for More Complex Neuron Network—Example 2:

A.



B. Error M:

$$\Rightarrow Error\,R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = Error\,R\cdot W_{OR}\cdot y_O'(z_O)$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = Error\,R\cdot W_{PR}\cdot y_P'(z_P)$$

$$\Rightarrow Error\,Q = \frac{\partial e}{\partial z_Q} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}$$

$$\Rightarrow Error\,Q = \frac{\partial e}{\partial z_Q} = Error\,R\cdot W_{QR}\cdot y_Q'(z_Q)$$

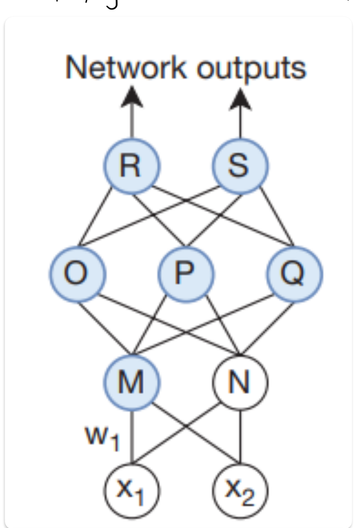
$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}\cdot\frac{\partial z_O}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}\cdot\frac{\partial z_P}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M}$$

$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = Error\,O\cdot W_{MO}\cdot f_M'(z_M) + Error\,P\cdot W_{MP}\cdot f_M'(z_M) + Error\,Q\cdot W_{MQ}\cdot f_M'(z_M) = (Error\,O\cdot W_{MO} + Error\,P\cdot W_{MP} + Error\,Q\cdot W_{MQ})\cdot f_M'(z_M)$$

C. $W_1' = W_1 + \eta\cdot(-1\cdot x_1\cdot Error\,M)$

3. Backpropagation for More Complex Neuron Network—Example 3:

A.



B. Error M:

$$\Rightarrow Error\,R = \frac{\partial e}{\partial z_R} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}$$

$$\Rightarrow Error\,S = \frac{\partial e}{\partial z_S} = \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}$$

$$\Rightarrow Error\,O = \frac{\partial e}{\partial z_O} = Error\,R\cdot W_{OR}\cdot y_O'(z_O) + Error\,S\cdot W_{OS}\cdot y_O'(z_O)$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}$$

$$\Rightarrow Error\,P = \frac{\partial e}{\partial z_P} = Error\,R\cdot W_{PR}\cdot y'(z_P) + Error\,S\cdot W_{PS}\cdot y_P'(z_P)$$

$$\Rightarrow Error\,Q = \frac{\partial e}{\partial z_Q} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}$$

$$\Rightarrow Error\,Q = \frac{\partial e}{\partial z_Q} = Error\,R\cdot W_{QR}\cdot y'(z_Q) + Error\,S\cdot W_{QS}\cdot y_Q'(z_Q)$$

$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}\cdot\frac{\partial z_O}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}\cdot\frac{\partial z_O}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}\cdot\frac{\partial z_P}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}\cdot\frac{\partial z_P}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}\cdot\frac{\partial z_Q}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M}$$

$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = \left(\frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_O}\cdot\frac{\partial y_O}{\partial z_O}\right)\cdot\frac{\partial z_O}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \left(\frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_P}\cdot\frac{\partial y_P}{\partial z_P}\right)\cdot\frac{\partial z_P}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M} + \left(\frac{\partial e}{\partial f_R}\cdot\frac{\partial f_R}{\partial z_R}\cdot\frac{\partial z_R}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q} + \frac{\partial e}{\partial f_S}\cdot\frac{\partial f_S}{\partial z_S}\cdot\frac{\partial z_S}{\partial y_Q}\cdot\frac{\partial y_Q}{\partial z_Q}\right)\cdot\frac{\partial z_Q}{\partial f_M}\cdot\frac{\partial f_M}{\partial z_M}$$

$$\Rightarrow Error\,M = \frac{\partial e}{\partial z_M} = (Error\,O\cdot W_{MO}\cdot f_M'(z_M) + Error\,P\cdot W_{MP}\cdot f_M'(z_M) + Error\,Q\cdot W_{MQ}\cdot f_M'(z_M) = (Error\,O\cdot W_{MO} + Error\,P\cdot W_{MP} + Error\,Q\cdot W_{MQ})\cdot f_M'(z_M)$$

C. $W_1' = W_1 + \eta\cdot(-1\cdot x_1\cdot Error\,M)$